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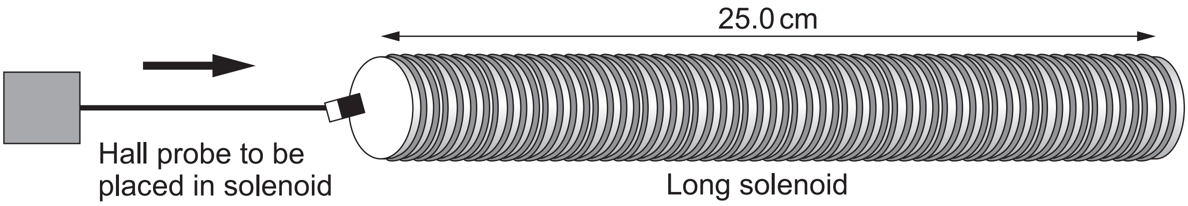
4. An experiment is carried out to measure the magnetic field in a long solenoid as the current in the solenoid is varied.



Theory states that the relationship between magnetic flux density, B, at the centre of the solenoid and current, I, is given by the equation:

$$B = \mu_0 n I$$

The results obtained are shown in the table and plotted on the grid opposite along with error bars and lines of maximum and minimum gradient.

Current / A ±0.01 A	Magnetic flux density / mT ±2 mT
0.50	11
1.00	20
1.50	29
2.00	41
2.50	49

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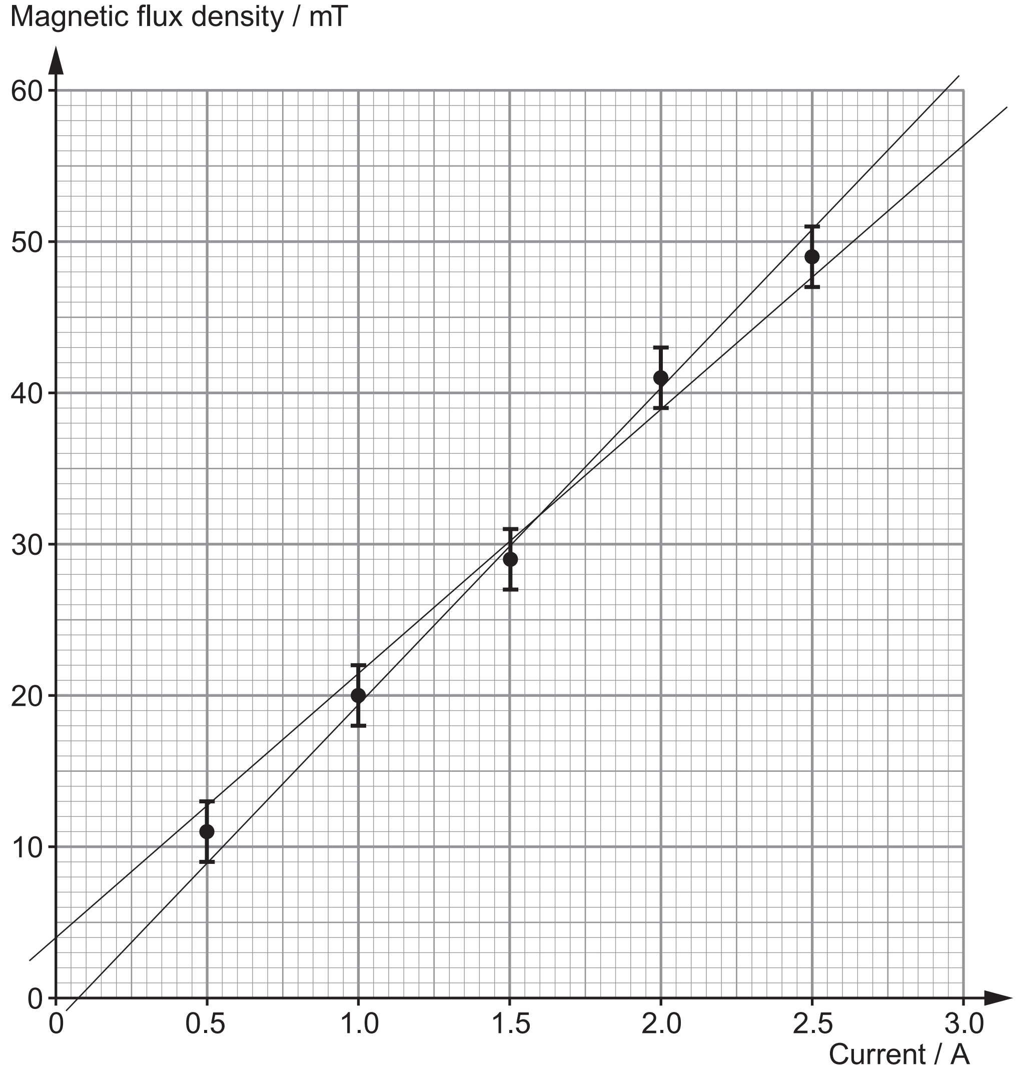

(a) Calculate the mean value of the gradient along with its **percentage** uncertainty. [4]

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(b) Calculate the number of turns in the solenoid along with its **absolute** uncertainty. The percentage uncertainty in the length of the solenoid is 0.4 %. [3]

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(c) (i) The solenoid manufacturer states that there are exactly 5 000 turns in the solenoid. Evaluate the accuracy of your value obtained in part *(b)* and whether or not the graph is in agreement with the equation: $B = \mu_0 nI$. [4]

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(ii) Suggest a reason for the disagreement between the manufacturer's stated value (5 000 turns) and your value calculated in part *(b)*. Suggest how the experimental technique might be improved for better agreement. [2]

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